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Methods and apparatus to secure communication devices to wristbands

Abstract

An example disclosed sheet includes a first layer; a second layer; a first die cut in the second layer defining a first wristband section and a second wristband section separated by a fold line; a second die cut in the first layer defining an imaging area; and a third die cut in the second layer defining an aperture in the second wristband section; wherein the aperture is surrounded by an adhesive and the aperture has an aperture dimension, the aperture dimension is greater than an inner dimension of the communication device, and the aperture dimension is less than an outer dimension of the communication device.

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Background/Summary

FIELD
(1) Examples disclosed herein are related to wristbands and, more particularly, methods and apparatus to secure communication devices to wristbands.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** depicts a first side of an example carrier containing an example wristband constructed in accordance with teachings of this disclosure.
(2) FIG. **2** depicts a second side of the example carrier of FIG. **1** opposite the first side.
(3) FIG. **3** depicts the example wristband of FIGS. **1** and **2** removed from the carrier.
(4) FIG. **4** depicts an example communication device constructed in accordance with teachings of this disclosure.
(5) FIG. **5** depicts the example wristband of FIGS. **1**, **2**, and **3** with the example communication device of FIG. **4** secured thereto by way of a lamination of the wristband.
(6) FIG. **6** is a view of a device receiving area of the wristband of FIGS. **1**, **2**, **4**, and **5**.
(7) FIG. **7** depicts another example carrier including another example wristband constructed in accordance with teachings of this disclosure.
(8) FIG. **8** depicts another example wristband constructed in accordance with teachings of this disclosure.
DETAILED DESCRIPTION
(9) Some wristbands are removably integrated into a carrier. The carrier is fed through a printer that is programmed to generate indicia on particular areas of the carrier, some of which correspond to

areas on the removable wristband. The indicia is, for example, human-readable text, a barcodes, or graphics. In addition to the printed indicia, some wristbands carry communication devices that store machine-readable data. For example, a passive radio frequency identification (RFID) tag having an identifier stored thereon may be permanently and fully embedded between layers (i.e., completely housed in and covered by layers) of the wristband. In such instances, the passive RFID tag includes a flat antenna implanted within the wristband.
(10) However, for certain applications or scenarios, passive RFID tags are not sufficient and other type(s) of communication devices are required. Passive RFID tags do not include an internal power source and, thus, rely on externally provided energy to operate. Specifically, passive RFID tags backscatter a signal provided by an external device (e.g., an RFID reader) to communicate data stored in a chip to the external device.
(11) Other types of communication devices, such as an active RFID tag, includes an internal power source and, thus, can operate without reliance on an external power source. Active RFID tags can be configured to transmit signals periodically using the internal power source. Because the active RFID tags do not rely on external device for energy, active RFID tags are preferred in some applications. For example, a hospital may want to continuously know the location of objects, such as patients, devices and/or inventory, throughout the hospital. Active RFID tags can be adhered to (or otherwise carried by) the objects to be located and configured to periodically transmit signals (e.g., beacons) to strategically placed RFID readers throughout the hospital. A locationing system including in communication with the RFID readers, which have known locations, receive the signals and calculate locations for the RFID tags based on the transmitted signals (e.g., using a triangulation technique and/or based on beacon).
(12) Examples disclosed herein provide a quick and secure attachment of a communication device, such as an active RFID tag, to an object, such as a wristband. While examples disclosed herein are described in connection with active RFID tags and wristbands, teachings of this disclosure are applicable to any suitable device for which attachment to any object is desired.
(13) An example wristband disclosed herein includes an aperture for receiving the communication device. The aperture of the example wristband is sized to allow an insertion portion of the communication device to pass through the aperture and to prevent a securing portion of the communication from passing through the aperture. Instead of passing through the aperture, the securing portion is positioned in a fold-over portion of a layer of the wristband. The fold-over portion of the wristband folds over and is adhered to itself to fasten the securing portion of the communication device within the fold-over portion (e.g., along with imaging areas of the wristband as described in some examples detailed below). With the communication device secured within the fold-over portion, the wristband can be attached to, for example, a person.
(14) As described in detail below, a user attaching the communication device to the wristband inserts the insertion portion of the communication device into the aperture and folds the fold-over portion of the wristband such that the material around the aperture is adhered to the area of the wristband including the securing portion of the communication device, thereby sealing the securing portion of the communication device between layers of the wristband.
(15) An example multi-layer carrier **102** including a wristband **100** constructed in accordance with teachings of this disclosure is shown in FIGS. **1** and **2**. The example depicted in FIG. **1** depicts the carrier **102** as a sheet, but the carrier **102** can be another other suitable type of carrier, such as a roll of wristbands (e.g., as described below in connection with FIG. **7**). Alternatively, the example wristband **100** can be provided separately as an individual unit (e.g., as described below in connection with FIG. **8**).
(16) The example carrier **102** of FIGS. **1** and **2** includes a first layer, a second layer, and adhesive between the first and second layers at certain locations on the carrier **102**. The view of FIG. **1** shows the first layer, which is sometimes referred to as a facesheet layer. In the example depicted in FIG. **1**, the first layer is paper. The first layer is capable of receiving printed indicia from, for

example, a media processing device (e.g., laser printer) programmed to generate the indicia at particular areas on the carrier **102**, including an imaging area **118** shown in FIG. **1**. In particular, the example carrier **102** of FIG. **1** is fed through the media processing device with the wristband **100** attached to the carrier **102**. Once the indicia is generated on, for example, the imaging area **118**, the wristband **100** is removed from the carrier **102**.

(17) The view of FIG. **2** shows the second layer. The second layer forms a side of the carrier **102** opposite the side formed by the first layer. In the illustrated example depicted in FIGS. **1** and **2**, the second layer is plasticine. In some embodiments, the second layer is a resilient material, such as plastic or nylon. In the example of FIGS. **1** and **2**, the second layer is translucent.

(18) A plurality of die cuts in the layers of the carrier **102** define the wristband **100** and certain areas of the wristband **100**, such as the imaging area **118**. The die cuts enable the wristband **100** to be removed from the carrier **102** and for portions of the first layer to separate from remaining portions of the first layer so that portions of the first layer that are adhered to corresponding portions of the second layer are separable from the carrier **100** together to form the wristband **100**. Specifically, some portions of the first layer are adhered to the second layer, such that the adhered portions of the first layer remain attached to the second layer upon removal of the wristband **100** from the carrier **102**.

(19) In the illustrated example, portions of the first layer corresponding to the imaging area **118**, an attachment patch **124**, and a placement area **116** are adhered to the second layer such that the imaging area **118**, the attachment patch **124**, and the placement area **116** are removed along with the second layer when the wristband **100** is detached from the carrier **102** (e.g., by pushing on the wristband **100**).

(20) As shown in the view of FIG. **2**, a first die cut **204** in the second layer defines an outline of the wristband **100**. That is, the first die cut **204** corresponds to an outer edge of the wristband **100** when detached from the carrier **102**.

(21) As shown in the view of FIG. **1**, a second die cut **122** in the first layer defines the imaging area **118** configured to receive printed indicia, such as a patient name or identifier. In the illustrated example, the second die cut **122** is a rectangle with rounded corners. Notably, the portion of the first layer positioned within the second die cut **122** is adhered to the corresponding portion of the second layer.

(22) As shown in the view of FIG. **1**, a third die cut **114** in the first layer defines the attachment patch **124**. The portion of the first layer within the third die cut **114** is releasably adhered (e.g., using a release layer) to the corresponding portion of the second layer, which includes of a layer of adhesive for use in securing the wristband **100** to a wearer.

(23) As show in the view of FIG. **1**, a fourth die cut **124** in the first layer defines the placement area **116**. As described in detail below, the placement area **116** is a location on the wristband **100** at which a communication device is attached to the wristband **100**. The portion of the first layer within the fourth die cut **124** is adhered to the corresponding portion of the second layer.

(24) Accordingly, when a force is applied within the outline formed by the first die cut **204**, the wristband **100** detaches from the carrier **102** and includes the portion of the second layer positioned with the first die cut **204**, the portion of the first layer corresponding to the imaging area **118**, the portion of the first layer corresponding to the attachment portion **114**, and the portion of the first layer corresponding to the placement area **116**.

(25) As shown in the view of FIG. **2**, a fifth die cut **202** in the second layer defines a device receiving aperture **206** configured to receive, for example, an RFID tag. Notably, the device receiving aperture **206** is not an aperture until the wristband **100** is removed from the carrier **102**. In the illustrated example, the portion of the second layer within the fifth die cut **202** is adhered to the corresponding portion of the first layer such that the portion of the second layer within the fifth die cut **202** remains on the carrier **102** when the wristband **100** is detached from the carrier **102**.

(26) FIG. **3** shows the wristband **100** removed from the carrier **102**. The wristband **100** has a first

wristband section **106** and a second wristband section **108** separated by a fold line **110**. The first wristband section **106** includes the imaging area **118** (formed by the first layer) and the placement area **116** (formed by the first layer). In the illustrated example, the imaging area **118** received printed indicia from a media processing device prior to the wristband **100** being detached from the carrier **102**. Example data to be imaged on the imaging area **118** includes a photographic image of the wearer, a barcode identifying the wearer, a trademark or logo, etc.

(27) The second wristband section **108** includes a lamination area **302** (formed by the second layer). The lamination area **302** is designed such that when the second wristband section **108** is folded atop the first wristband section **106** along the fold line **110**, the lamination area **302** covers and protects (e.g., laminates) the imaging area **118**. In the illustrated example, the lamination area **302** is translucent to allow viewing of the imaging area **118** and the indicia printed thereon. At least a portion of the lamination area **302** is covered by adhesive. The portion of the lamination area **302** containing the adhesive may be a small percentage of the lamination area **302** (e.g., an outline along the outer edge) or it may be the entire lamination area **302**. The adhesive on the lamination area **302** adheres the first wristband section **106** to the second wristband section **108** to place the wristband **100** in a state ready for attaching to a wearer (as shown in FIG. **5**).

(28) The lamination area **302** includes the device receiving aperture **206** having an aperture dimension **306**. Although the aperture **206** is depicted in the illustrated example as a circle to receive a circular communication device, the aperture **206** may be any suitable shape that is correspondingly shaped with the communication device to be received therein. That is, the aperture **206** can be of any shape to accommodate alternatively shaped communication devices. As the aperture **206** illustrated in the example of FIG. **3** is circular, the aperture dimension **306** is a diameter. Alternatively, if the aperture **206** was, for example, a rectangle, the aperture dimension **306** would be, for example, a width or length or combination thereof.

(29) In the depicted embodiment, the aperture **206** is positioned on the second wristband section **108** across the fold line **110** from the placement area **116** at which a communication device, such as an RFID tag is placed for securing to the wristband **100**. Thus, when the second wristband section **108** is folded atop the first wristband section **106**, the aperture **206** aligns with the placement area **116**. This configuration results an aperture-shaped portion of the first wristband section **106** corresponding to the placement area **116** not being laminated and the portion surrounding the placement area **116** being laminated.

(30) As further described below, by placing a communication device on the placement area **116**, the aperture **206** folds over the communication device, allowing one portion of the communication device to pass through the aperture **206** and securing another portion of the communication device between the first and second wristband sections **106** and **108**.

(31) FIG. **4** shows an example communication device **400** constructed in accordance with teachings of this disclosure. The example communication device **400** includes an insertion portion **402** having an inner dimension **406** and a securing portion **404** having an outer dimension **408**. As the example communication device **400** of FIG. **4** is circular, the inner dimension **406** and the outer dimension **408** are diameters and are sometimes referred to herein as diameters. However, differently shaped communication devices may have different dimensions, such as inner and outer widths, or inner and outer lengths.

(32) The communication device **400** is, for example, a radio frequency identification tag, a Bluetooth beacon, an ultra-wide band (UWB) tag, a near field communication (NFC) tag, or any other suitable type of device. As depicted in FIG. **4**, the example communication device is an RFID tag. In some embodiments, the securing portion **404** and the insertion portion **402** are part of the same structure (i.e., are integrally formed). In some embodiments, the securing portion **404** is a flange attached to a surface of the insertion portion **402**. The securing portion **404** is, for example, plastic, paper, fiberglass, laminate, or any other material that is resilient such that retention of the securing portion **404** results in the communication device **400** being secured to the wristband **100**.

(33) The example communication device **400** as shown in FIG. **4** is an active RFID tag. As is noted above, active RFID tags require a larger housing relative to passive RFID tags, to house a battery, an antenna, circuitry, etc. Due to the larger housing, active RFID tags cannot be sealed flat between the first and second portion of the wristband and must be retained in some other way (as addressed by teachings of this disclosure). The aperture **206** allows for the larger housing of active RFID tags to protrude while other portions of the active RFID tag are retained.
(34) Notably, the outer diameter **408** is greater than the inner diameter **406**. The difference in the outer diameter **406** and the inner diameter **408** allows the insertion portion **402** to pass through the aperture **206** while the securing portion **404** is unable to pass through the aperture **206**. The aperture diameter **306** is greater than the inner diameter **406**, and the aperture diameter **306** is lesser than the outer diameter **408**. This allows the aperture **206** to pass over the insertion portion **402** of the communication device **400** with the communication device **400** positioned on the placement area **116**, and the area of the lamination area **302** surrounding the aperture **206** to contact the second wristband section **106**, thereby trapping the securing portion **404** in between the first and second wristband sections **106** and **108**, which will secure the communication device **400** to the wristband **100**.
(35) In other examples configured for differently shaped communication devices, such as a rectangular RFID tag, the outer width or length is less than the inner width or length to enable the insertion portion of the communication device to pass through the aperture but not the securing portion. That is, alternatively shaped apertures are sized correspondingly to the shape of communication device to be received therethrough.
(36) FIG. **5** shows the wristband **100** folded along the fold line **110** such that the first wristband section **106** and the second wristband section **108** are overlaid and the communication device **400** is secured therebetween. When the second wristband section **108** overlays the first wristband section **106**, the lamination area **302** laminates the imaging area **118**. The lamination protects the imaging area **118** such that any indicia printed thereon will not be directly smudged, scratched, dissolved, or otherwise damaged. When the second wristband section **108** is folded over the first wristband section **106**, the aperture **206** aligns with the placement area **116**. As is depicted in FIG. **5**, when the communication device **400** is positioned on the placement area **116**, the second wristband section **108** is folded over the first wristband section **106** along the fold line **110**, the aperture **206** passes over the communication device **400**.
(37) An enhanced view of box F.**6** around the communication device **400** is shown in FIG. **6**. That is, FIG. **6** depicts a close-up of the communication device **400** and the aperture **206** of FIG. **5**. In FIG. **6**, the surfaces of the communication device **400** can be seen, which includes the insertion portion **402** and the securing portion **404**. As shown in FIG. **6**, the insertion portion **402** is small enough to fit through aperture **206**, however the securing portion **404** is not. Therefore, when the second wristband section **108** is folded over to laminate the first wristband section **106**, the lamination area **302** retains the securing portion **404** of the communication device **400**. The lamination area **302** features adhesive which allows the lamination area **302** to adhere to surfaces that it contacts. In the example of FIG. **6**, the lamination area **302** adheres to areas surrounding (and perhaps including) the placement area **116**.
(38) As depicted in FIG. **6**, an overlap area **600** of the lamination area **302** is where the lamination area **302** contacts the securing portion **404** of the communication device **400**. The lamination area **302** retains the securing portion **404** in part due to the contact in the overlap area **600**. In some embodiments, the lamination area **302** in the overlap area **600** has adhesive. As the lamination area **302** has contact with the securing portion **404** and the areas of the lamination area **302** outside of the overlap area **600** are adhering to the first wristband section **106**, the securing portion **604** will be secured between the lamination area **302** and the placement area **116**.
(39) FIG. **7** illustrates another example constructed in accordance with teachings of this disclosure. The example of FIG. **7** is another example carrier, roll **700** that contains a chain of carriers **702**. In

FIG. **7**, each carrier **702** is removeable from the roll **700** via perforations **704** between the individual carriers **702**. In some embodiments, the carrier **702** is processed by a media processing device prior to separating from the roll **700**. While the carrier **102** described above is capable of being processed by a laser printer, the roll **700** is capable of being processed by a thermal printer. Each carrier **702** functions similar to the two-layered carrier **102** described above. A wristband is diecut within the carrier **702** such that the wristband can be removed from the carrier **702**. Each wristband contains an aperture similar to the aperture **206** discussed above. The wristband has a fold line, where opposite the fold line is adapted to receive a communication device. An imaging area receives indicia to be generated on the wristband by, for example, a thermal printer. When the wristband is folded over the fold line **710**, the aperture may pass over and secure a communication device as described above.

(40) FIG. **8** depicts another example wristband **800** constructed in accordance with teachings of this disclosure. The example wristband **800** of FIG. **8** includes a strap **802** and a tab **804** extending from a side edge of the strap **802**. The tab **804** may be located at any point along a length of the strap **802**, and the tab **804** may have a width up to the length of the strap **802**.

(41) The tab **804** includes an aperture **806**. In the illustrated example, the aperture **806** is completely contained within the tab **804**. In the depicted embodiment, the surface of the tab **804** around the aperture **806** include adhesive, such that when the tab **804** is folded over onto the strap **802**, the adhesive would adhere the tab **804** to the strap **802**.

(42) Like the aperture **206** of FIG. **6**, the aperture **806** of FIG. **8** is sized to receive a communication device. The communication device is capable of being placed on the strap **802** opposite from the tab **804**, the communication device then being inserted through the aperture **806** such that the aperture **806** fits over the communication device and secures the communication device to the strap **802**.

(43) The wristband **802** has a securing mechanism **808** at an end. FIG. **8** depicts the securing mechanism as an adhesive capable of adhering opposite ends of the strap **802** together. The securing mechanism **808** can be any means for securing one end of the strap **802** to the other end to loop the wristband **800** around a wearer's wrist. Other wristband embodiments may include securing mechanisms featuring a snap and peg feature, a locking feature, or other similar securing method known for wristbands.

(44) The illustrations described herein are intended to provide a general understanding of the structure of various embodiments. The illustrations are not intended to serve as a complete description of all of the elements and features of apparatus that utilize the structures or methods described herein. Many other embodiments may be apparent to those of skill in the art upon reviewing the disclosure. Other embodiments may be utilized and derived from the disclosure, such that structural and logical substitutions and changes may be made without departing from the scope of the disclosure. Additionally, the illustrations are merely representational and may not be drawn to scale. Certain proportions within the illustrations may be exaggerated, while other proportions may be minimized. Accordingly, the disclosure and the figures are to be regarded as illustrative rather than restrictive.

(45) The above disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other embodiments, which fall within the true spirit and scope of the description. Thus, to the maximum extent allowed by law, the scope is to be determined by the broadest permissible interpretation of the following claims and their equivalents and shall not be restricted or limited by the foregoing detailed description.

Claims

1. A carrier including a detachable wristband, the carrier comprising: first and second wristband section defined by a first die cut; a fold line between the first and second wristband sections, the first wristband section and the second wristband section joined by and asymmetrically disposed about a fold line an imaging area defined by a second die cut in the first wristband section adjacent to a placement area; and an aperture defined by a third die cut in the second wristband section, wherein when the second wristband section is folded over the fold line, the second wristband section overlays the first wrist band section and the aperture aligns with and overlays the placement area and is configured and dimensioned to retain a first portion of a radiofrequency device against the placement area while the aperture is configured and dimensioned to allow a second portion of the radiofrequency device to extend through the aperture.
2. The carrier of claim 1, wherein, when the communication device is positioned on a placement area located on the first wristband section and the second wristband section is folded along the fold line, an insertion portion of the communication device passes through the aperture and a securing portion of the communication device is secured between the first and second wristband sections.
3. The carrier of claim 1, wherein, when the first wristband section is folded along the fold line and brought into contact with the first wristband section, the first and second wristband sections are adhered together.
4. The carrier of claim 1, wherein the first wristband section has a first attachment section and the second wristband section has a second attachment section, and when the wristband forms a closed loop when the first attachment section contacts the second attachment section.
5. The carrier of claim 1, wherein the wireless communication device is an active radio frequency identification tag.
6. The carrier of claim 1, wherein the wireless communication device is a Bluetooth beacon.
7. The carrier of claim 1, wherein the insertion portion and the securing portion are integrally formed.
8. The carrier of claim 1, wherein the aperture is shaped correspondingly with the communication device.
9. The carrier of claim 1, wherein the aperture, inner, and outer dimensions are diameters.
10. A wristband comprising: a first wristband section including an imaging area configured to receive indicia and a placement area to receive a first portion of a radiofrequency device; a second wristband section joined to the first wristband section by and asymmetrically disposed about a fold line; and an aperture formed in the second wristband section, wherein when the second wristband section is folded over the fold line, the second wristband section overlays the first wrist band section and the aperture aligns with and overlays the placement area and is configured and dimensioned to retain a first portion of a radiofrequency device against the placement area while the aperture is configured and dimensioned to allow a second portion of the radiofrequency device to extend through the aperture.
11. The wristband of claim 10, wherein the aperture is circular.
12. The wristband of claim 10, wherein the communication device is an active radio frequency identification tag.
13. The wristband of claim 10, wherein the first wristband section has a first attachment section and the second wristband section has a second attachment section, wherein when the wristband forms a closed loop the first attachment section contacts the second attachment section.
14. The wristband of claim 10 wherein the imaging area is part of a first layer and the first and second wristbands sections are part of a second layer.
15. The wristband of claim 10, wherein the first layer is a first layer and the second layer is a second layer.
16. The wristband of claim 10, wherein the aperture, inner, and outer dimensions are diameters.